

Real Business Cycle (RBC) Model

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Introduction

- First RBC model dates back to Kydland and Prescott (1982)
 - Added complications to increase persistence to shocks
- Hansen (1986) developed a basic RBC model with essential ingredients
- In RBC world, supply shocks generate economic fluctuations
- Neoclassical growth model as reference for economy's long-term behavior
 - Perfect competition and flexible prices
- Model describes behavior of two types of **representative agents**
 - A large number of identical consumers
 - A large number of identical firms

Model Structure

- Assumptions about economy and agents
- Agents' optimization problems
 - Households (HHs)
 - Firms
- Model's equilibrium conditions
- Steady state
- Calibration
- Log-linearization
- Dynare implementation
- Impulse response functions

Assumptions: Economy

- Closed economy (no imports, no exports)
- Only one (physical) good (consumed or used as capital)
- No financial sector (interest rate is a proxy for return on capital)
- No government (no taxes, no central bank)
- No adjustment costs (for changes in capital stock)
- No population growth

Assumptions: Households

- Continuum of mass 1 of identical HHs indexed by $i \in [0, 1]$
 - Dividing aggregate variable by 1 gives the per capita
 - Remove index i to simplify notation
- HHs are owners of factors of production: Capital and labor
 - Factor markets are perfectly competitive (no rigidities)
- Each HH maximizes an intertemporal utility function
 - HH decides how much to consume, work and save (investment in capital)
- Utility is intertemporally additively separable (no habit formation) into:
 - Consumption (utility) and labor (disutility)

Assumptions: Households (Preferences)

- Optimization problems have well-defined solutions when objective function is concave and solution spaces are convex sets
- Preferences represented by a utility function: $U(C_t, H_t)$
 - $U_C > 0$ (utility); $U_H < 0$ (disutility); $U_{CC} < 0, U_{HH} < 0$ (concave)
 - Commonly used utility functions: Logarithm, CRRA, CES
- Future utility at present using an intertemporal discount **factor**: β
 - $\beta = 1/(1 + \tau)$, where τ is the subjective **rate** of time preference
- Expectations operator conditional on time t information: $\mathbb{E}_t(\cdot) = \mathbb{E}(\cdot|I_t)$
- Each period capital depreciates at rate: δ

Assumptions: Firms

- Continuum of mass 1 of identical firms indexed by $i \in [0, 1]$
 - Remove index i to simplify notation
- Each firm maximizes profits under perfect competition
- Technology represented by a production function: $F(K_t, H_t)$
 - $F_K > 0, F_H > 0$ (increasing); $F_{KK} < 0, F_{HH} < 0$ (concave)
 - Constant returns to scale: $F(zK_t, zH_t) = zY_t$
 - Inada conditions:
 $\lim_{K \rightarrow 0} F_K = \infty, \lim_{K \rightarrow \infty} F_K = 0, \lim_{H \rightarrow 0} F_H = \infty, \lim_{H \rightarrow \infty} F_H = 0$
 - Commonly used production functions: AK , Cobb-Douglas, CES

Handout

- Agents' optimization problems
 - Households
 - Firms
- First order conditions
- Model's equilibrium conditions
- Steady state
- Calibration
- Log-linearization
- Dynare implementation

Variables: Household

- C_t : Units of the consumption good
- H_t : Hours dedicated to labor
- K_t : Units of the good representing the capital stock
- I_t : Units of the good invested as capital
- T_t : Dividends or profits of the firm
- w_t : Wage per hour of labor
- r_t : Return per unit of capital
- p_t : Aggregate price level
- $\beta, \gamma, \varphi, \delta$: Parameters

Variables: Firm

- Y_t : Output
- A_t : Productivity
- V_t : Revenue
- TC_t : Total cost
- MC_t : Marginal cost
- T_t : Profits of the firm
- α, ρ_A, σ_A : Parameters

Technical: Unique and Stable Equilibrium

- A non-linear model can be linearized
- Model in linearized form can be expressed as

$$A \begin{bmatrix} z_t \\ E_t x_{t+1} \end{bmatrix} = B \begin{bmatrix} z_{t-1} \\ x_t \end{bmatrix} + C \varepsilon_t$$

where:

- z_t is a vector of predetermined variables at time t
- x_t is a vector of forward-looking variables
- ε_t is a vector of shocks
- A , B and C are matrices

Technical: Unique and Stable Equilibrium

- Blanchard and Kahn (1980): Unique equilibrium and stability condition depends on magnitude of eigenvalues (λ) of matrix B
 - Eigenvalues that satisfy $|\lambda| > 1$ are called unstable roots
 - Unstable roots arise in dynamic models with forward-looking variables
- Condition: If number of unstable roots equals number of forward-looking variables, system has a **unique** solution and a **stable** saddle path
- Otherwise, indeterminacy: Multiple solutions or no solution
 - Causes: Calibration errors, model misspecification, ill-behaved expectations

Matlab

- Programming language developed by MathWorks
 - GUI: Command Window, Editor, Workspace, Current Folder
 - [Link](#) to quick reference guide
- Mainly used in engineering and economics for research and development
- Built around matrices and arrays, so it is easy to operate with them
- Variable names must start with a letter; followed by letters, digits or `_`
- Matlab (and Dynare) are case sensitive: $x \neq X$
 - Beware: Some commands are Dynare-specific
- Write code in `.m` file and run it by typing its name in command window
 - `.mat` files store workspace variables

Dynare

- Install Dynare from <http://www.dynare.org>
- Every time you start Matlab, add Dynare folder to Matlab path
 - Type: `addpath C:\dynare\x.y\matlab` (x.y is the version)
 - Set Path → [dynare installation directory] → matlab → Add Folder...
- Write your model in a text file and save it with extension .mod
 - Preamble (define variables)
 - Model (and steady state) block(s)
 - Shocks
 - Computations
- In Matlab, make sure current folder is where your .mod file is located
- Run it by typing `dynare filename.mod` in command window

.mod Files

- All code lines must end with a semicolon ;
- Comments can be made with `//` or `%` for lines, `/* */` for sections of code
- Blocks (model, steady_state_model, shocks) closed using `end;` command
- Equations entered one after another, each one ended with ;
- Model needs to have as many equations as endogenous variables
- Variable and parameter names are case sensitive, must match preamble
- Default timing is t, so x_t is written as `x`
 - Use `x(+1)` for leads and `x(-1)` for lags (valid for any lead or lag)
- No need to enter expectation operator, Dynare adds it automatically

Dynare Output

- Impulse response functions (IRFs)
 - What to look for? Scale, on impact effect relative to zero, periods, shape
 - No confidence bands if model is calibrated (no uncertainty)
- All of Dynare output is stored in data structures, most important is `oo_`
- Information about model can be found in structure `M_`
- Linearization can be done manually
 - Equations in model block already linearized
 - In linearized model, all variables have zero steady state
- Dynare can also solve non-linear models (approx. to higher orders)
 - Simulations use extended path methods